# Jahanvi Khambalkar

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## EDUCATION

**Master of Professional Studies in Data Science** **Aug 2022 - May 2024**

- University of Maryland, Baltimore County Baltimore, Maryland
- Coursework – Data Analysis, Machine Learning, AI, Database Management, Big Data Processing, (Financial) Data Science

**Bachelor of Engineering in Computer Engineering** **Aug 2017 - May 2021**

- L.D.R.P. Institute of Technology and Research Gujarat, India

## TECHNICAL SKILLS

- **Programming Languages:** Python, JavaScript, Java, C, C++, CSS
- **Libraries/Frameworks:** Pandas, NumPy, Matplotlib, Scikit-Learn, Seaborn, TensorFlow, Keras, PyTorch, OpenCV, Bootstrap
- **Web Technologies:** HTML, React JS, Node JS, REST APIs, JSON
- **Version Control and Tools:** Git, GitHub, Gitlab

## EXPERIENCE

**Graduate Teaching Assistant | Department of Data Science, UMBC** **Jan 2023 - Dec 2023**

- Held office hours for "DATA605 - Ethical and Legal Issues in Data Science" and assisted by addressing and elucidating coursework queries, providing clarification on complex concepts, and offering support in writing papers.
- Evaluated assignments and discussion forums for around 50 students across two classes, delivering actionable feedback and upholding stringent standards for fairness, accuracy, and academic integrity.
- Monitored and addressed concerns related to plagiarism or unethical practices.

**Web Developer Intern | Nimblechapps Pvt. Ltd.** **April 2021 - Oct 2021**

- Created web applications with ReactJS and NodeJS, designing UI and backend services to streamline business processes.
- Engineered solutions using a test-driven approach, incorporating robust security protocols, data protection measures, and efficient storage solutions.

**Data Science & Business Analyst Intern | The Sparks Foundation** **Jan 2021 - Apr 2021**

- Spearheaded comprehensive Exploratory Data Analysis techniques to extract actionable insights, strategically addressing intricate business problems and fostering data-driven decision-making.
- Trained and Implemented Machine Learning models to predict key business outcomes and optimize revenue generation.
- Identified critical areas for improvement through comprehensive analysis and proposed data-driven solutions that directly influenced business decisions.

## PROJECTS

**Time Series Analysis to Forecast Bitcoin Price | Financial Machine Learning** **Dec 2023**

- Conducted EDA, analyzed components of time series(trends, seasonality, and residuals), implemented the Augmented Dicky Fuller test to assess stationarity, and applied differencing to make the data stationary.
- Determined model parameters (p, d, q) using ACF and PACF Plots. Explored and compared different models, including Auto-Regressive, Moving Average, Auto-ARIMA, and traditional ARIMA, selecting the best-performing models based on Root Mean Squared Error (RMSE) for forecasting Bitcoin prices.

**Customer Segmentation using RFM Analysis | Unsupervised Machine Learning** **Dec 2023**

- Processed an unlabeled dataset from the UCI Machine Learning Repository, generating Recency, Frequency, and Monetary features for each customer using custom functions. Utilized K-Mean clustering on the scaled data to identify distinct customer clusters based on RFM features.
- Determined Min and Max Threshold for R, F, M features by analyzing the data distribution and segmented each cluster further based on feature variation within the cluster, using the derived thresholds.
- Evaluated clustering performance using metrics like the Calinski-Harabasz Index, Silhouette Score, and Davies-Bouldin Index. Developed a Streamlit application to visually demonstrate the working of the model.

**Facial Emotion Detection | Deep Learning** **Dec 2022**

- Built a Flask web application incorporating deep neural networks to classify facial expressions from images or videos.
- Introduced continuous time-based emotional dimensions (e.g., arousal, valence) to analyze both discrete and continuous shifts in expressions, thereby enriching the representation of the full spectrum of emotions.

**Personal Desktop Assistant | Python** **Dec 2021**

- Developed a Python-based assistant capable of executing various tasks in response to voice commands of users, such as playing music, launching applications, and providing information.
- Formulated responses using a text-to-speech feature and incorporated features for sending/scheduling emails and WhatsApp texts over voice commands.